

Module 2 Supervised Machine Learning

Decision Tree – ID3

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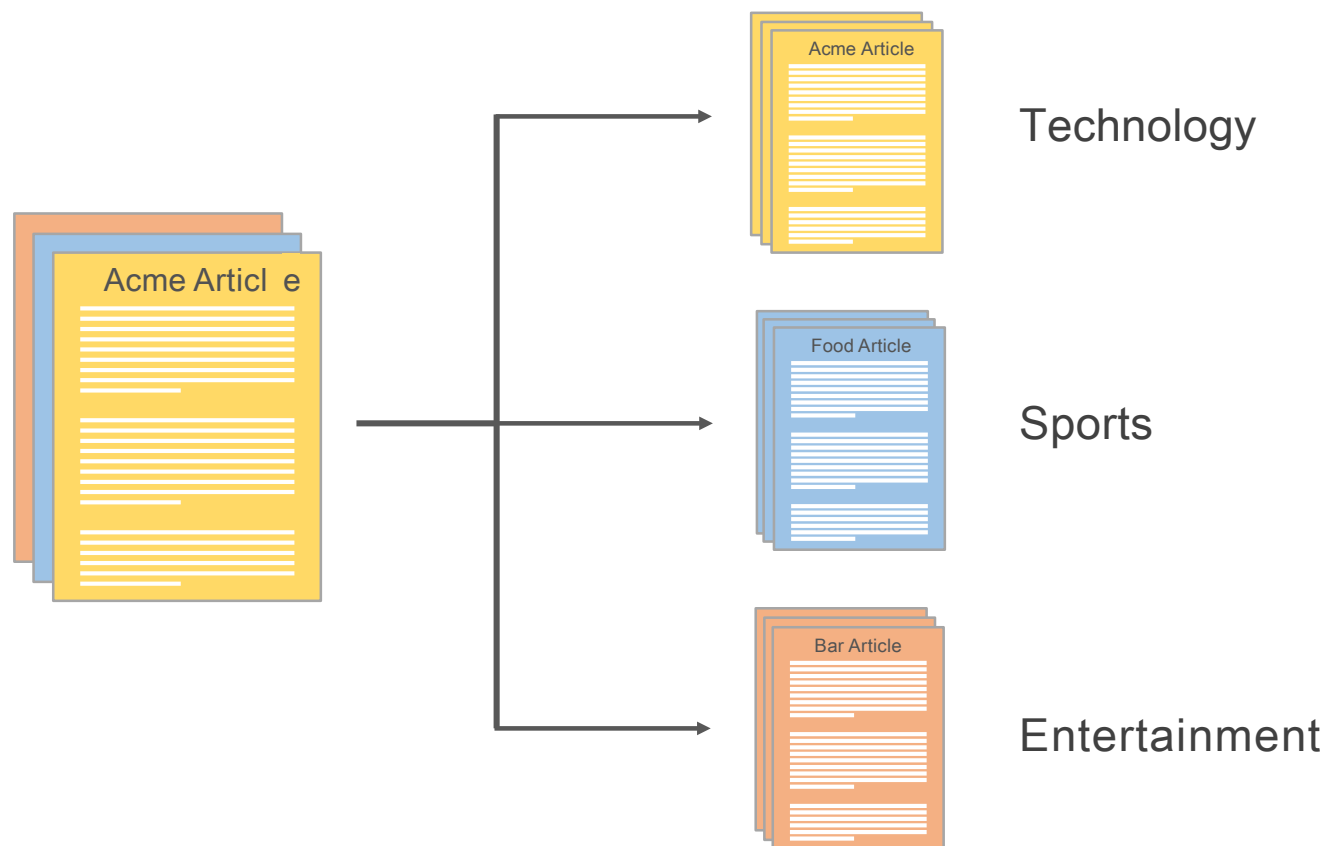
Department of Computer Engineering KJSSE

Outline

- Decision Trees: Structure of decision trees,
- Tree building algorithms (e.g., ID3, CART, C4.5),
- Ensemble learning, bagging and Random Forests.

What Is Classification?

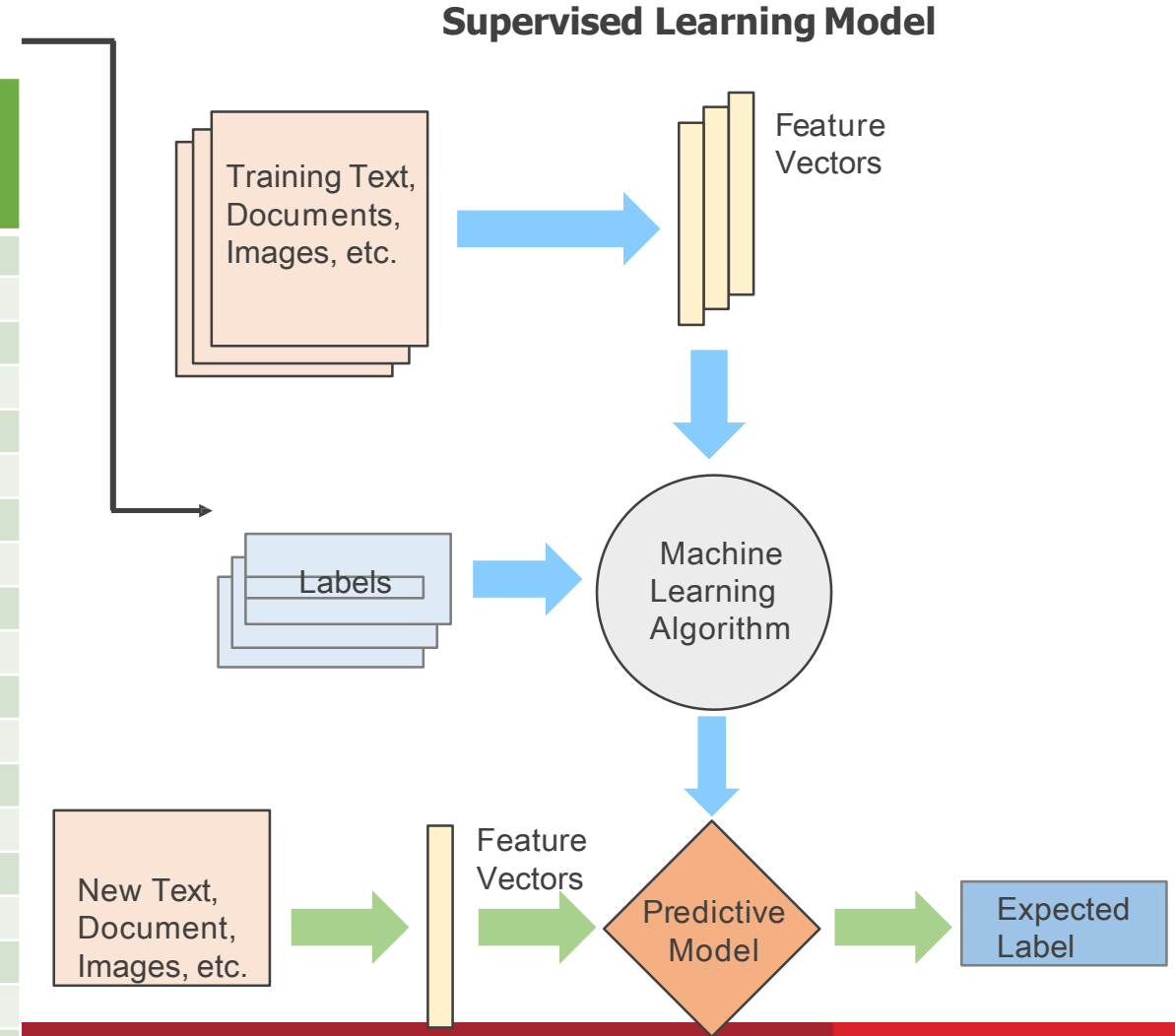
A machine learning task that identifies the class to which an instance belongs



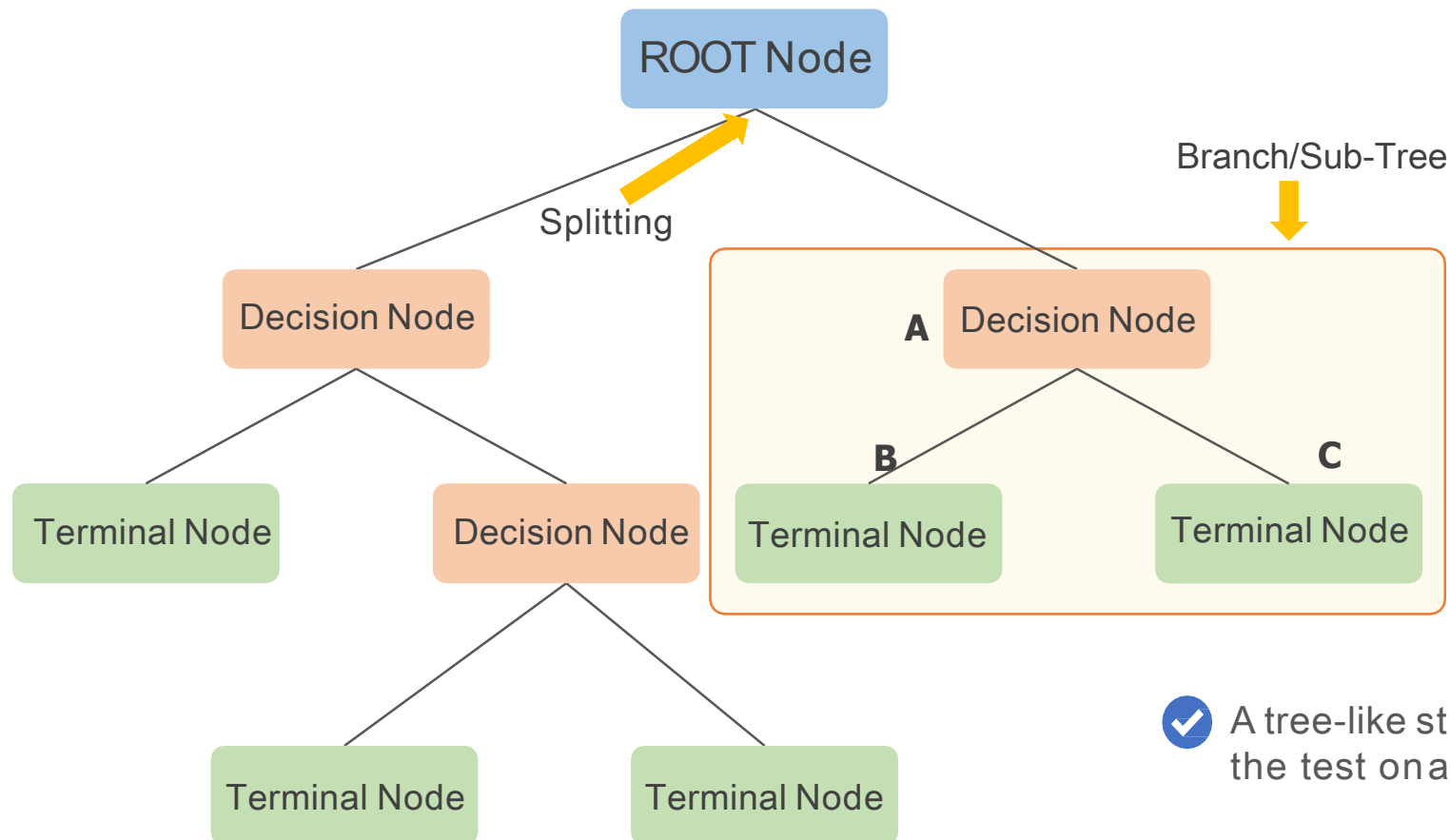
Classification: A Supervised Learning Algorithm

Classification is a supervised learning algorithm as the training data contains labels

Record ID	Age	Spectacle Prescription	Astigmatic	Tear production Rate	Class Label Lenses
1	Young	Myope	No	Reduced	Noncontact
2	Young	Myope	No	Normal	Soft contact
3	Young	Myope	Yes	Reduced	Noncontact
4	Young	Myope	Yes	Normal	Hard contact
5	Young	Hypermetrope	No	Reduced	Noncontact
6	Young	Hypermetrope	No	Normal	Soft contact
7	Young	Hypermetrope	Yes	Reduced	Noncontact
8	Young	Hypermetrope	Yes	Normal	Hard contact
9	Pre-presbyopic	Myope	No	Reduced	Noncontact
10	Pre-presbyopic	Myope	No	Normal	Soft contact
11	Pre-presbyopic	Myope	Yes	Reduced	Noncontact
12	Pre-presbyopic	Myope	Yes	Normal	Hard contact
13	Pre-presbyopic	Hypermetrope	No	Reduced	Noncontact
14	Pre-presbyopic	Hypermetrope	No	Normal	Soft contact
15	Pre-presbyopic	Hypermetrope	Yes	Reduced	Noncontact
16	Pre-presbyopic	Hypermetrope	Yes	Normal	Noncontact
17	Presbyopic	Myope	No	Reduced	Noncontact
18	Presbyopic	Myope	No	Normal	Noncontact
19	Presbyopic	Myope	Yes	Reduced	Noncontact
20	Presbyopic	Myope	Yes	Normal	Hard contact



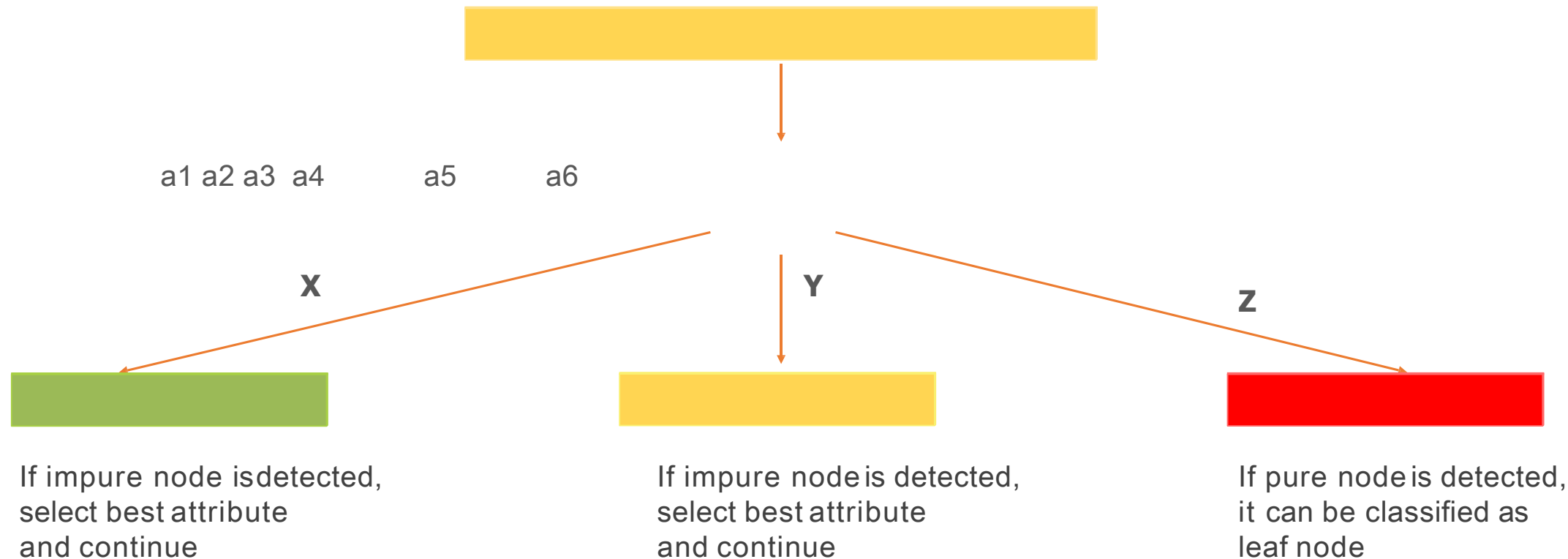
Decision Tree Classifier



- ✓ A tree-like structure in which the internal node represents the test on an attribute
- ✓ Each branch represents the outcome of the test, and each leaf node represents the class label
- ✓ A path from root to leaf represents classification rules

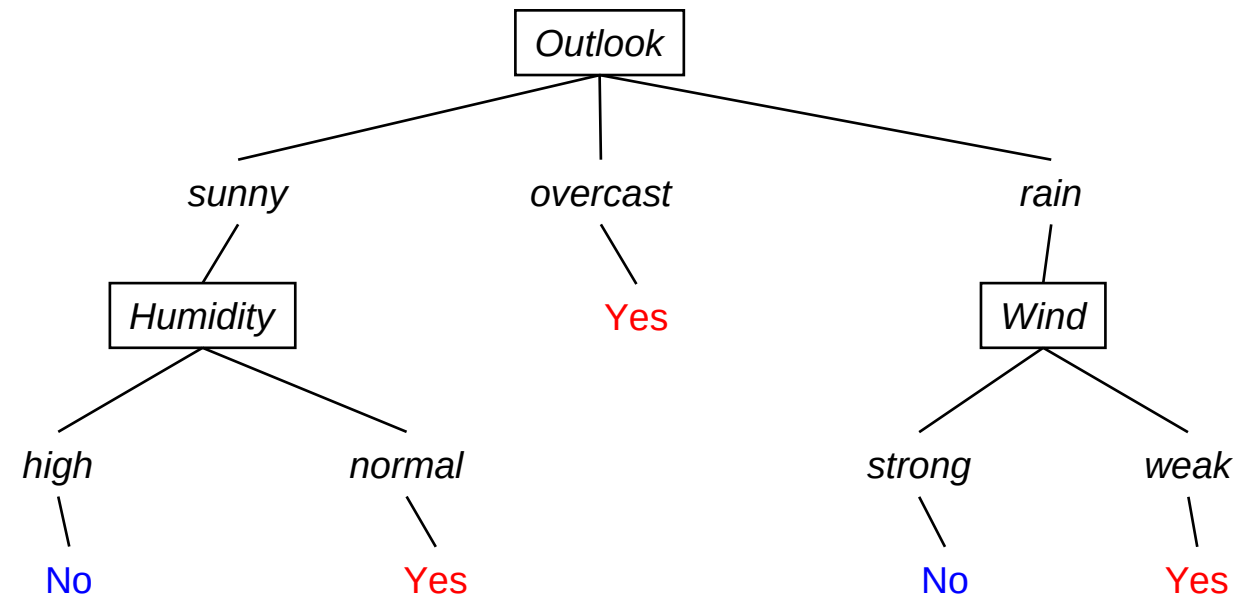
Decision Tree: Schematic Representation

The tree is splitted whenever an impure node is detected



Decision Trees

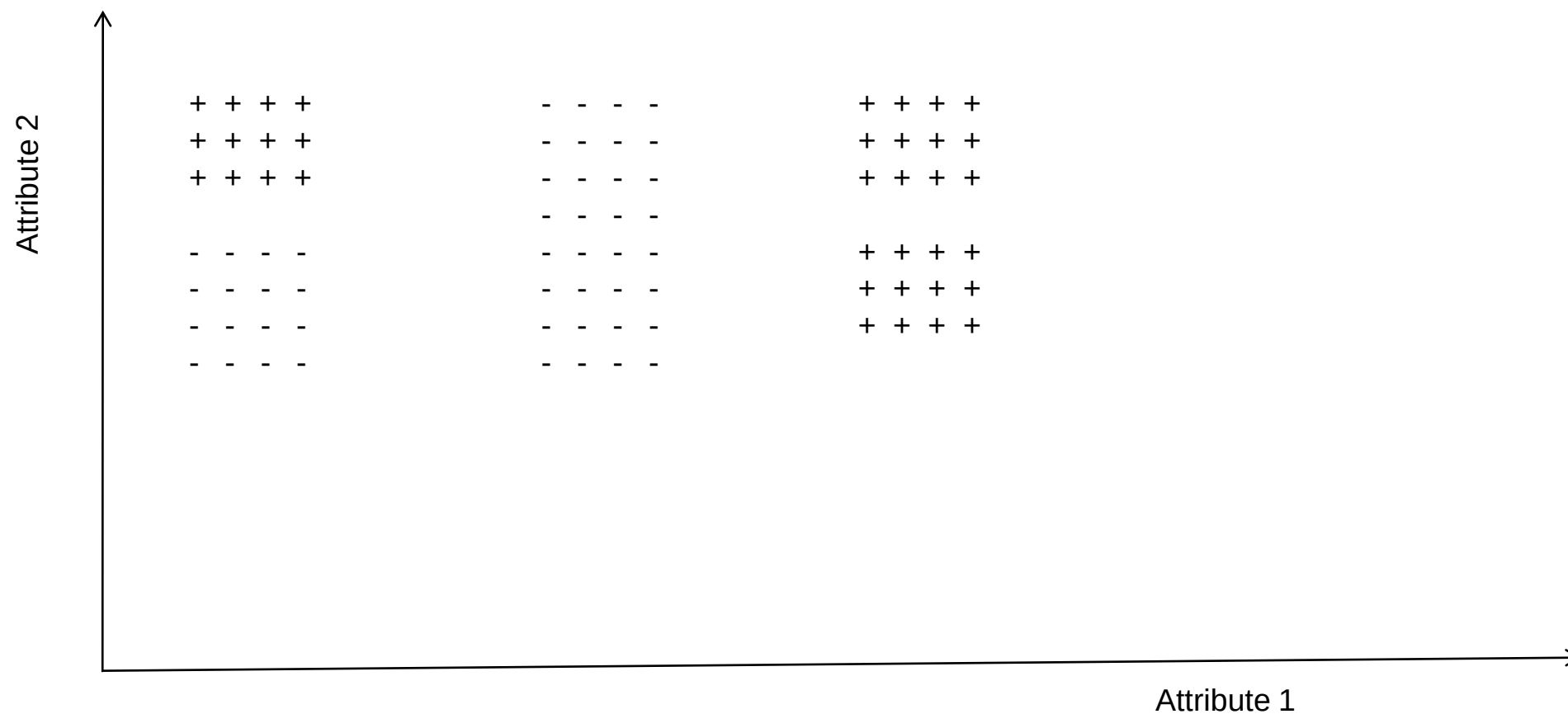
- Decision tree to represent learned target functions
 - Each internal node tests an attribute
 - Each branch corresponds to attribute value
 - Each leaf node assigns a classification
- Can be represented by logical formulas



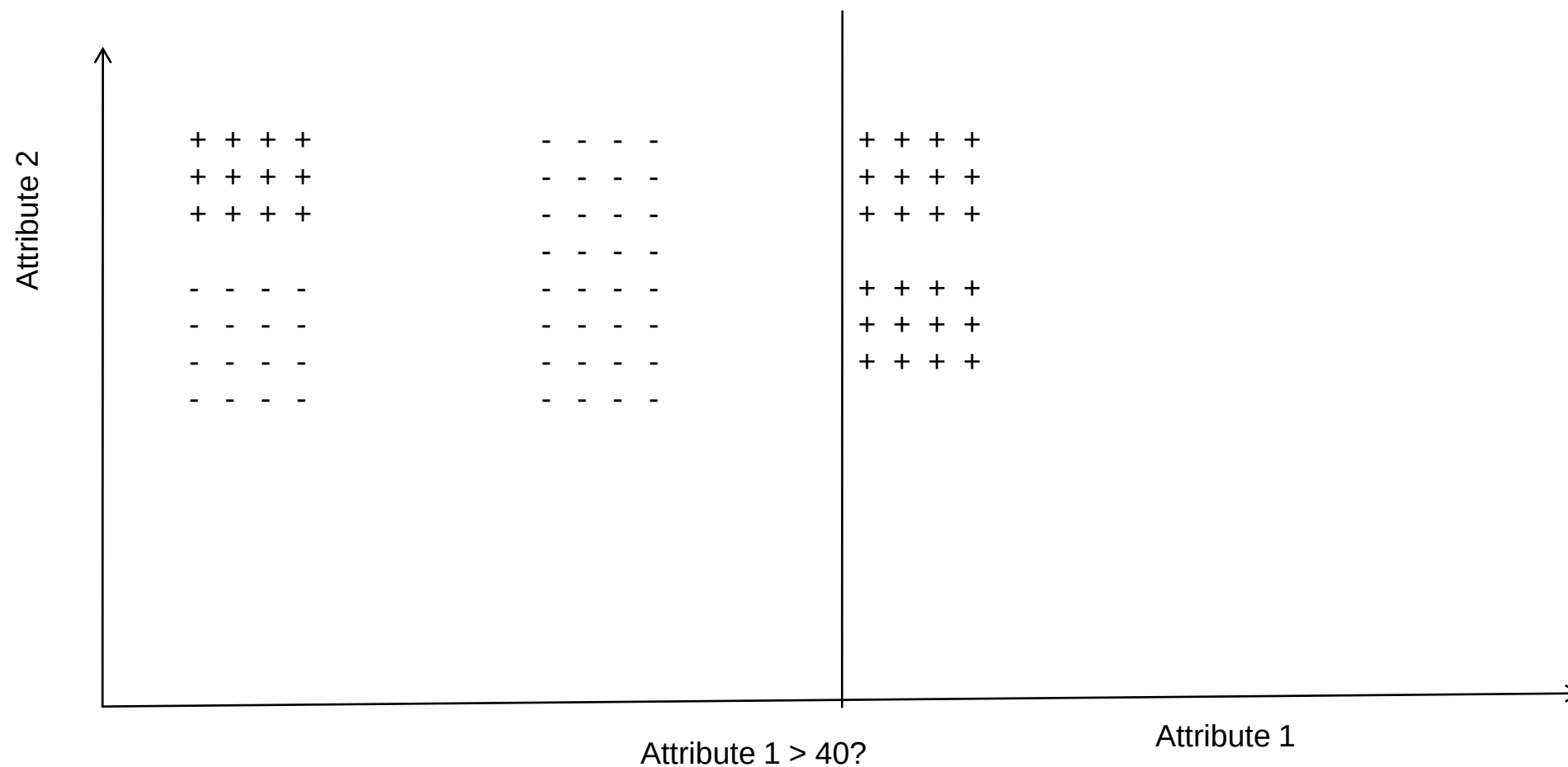
Top-Down Construction

- Start with empty tree
- Main loop:
 1. Split the “best” decision attribute (A) for next node
 2. Assign A as decision attribute for node
 3. For each value of A, create new descendant of node
 4. Sort training examples to leaf nodes
 5. If training examples perfectly classified, STOP,
Else iterate over new leaf nodes
- Grow tree just deep enough for perfect classification
 - If possible (or can approximate at chosen depth)
- Which attribute is best?

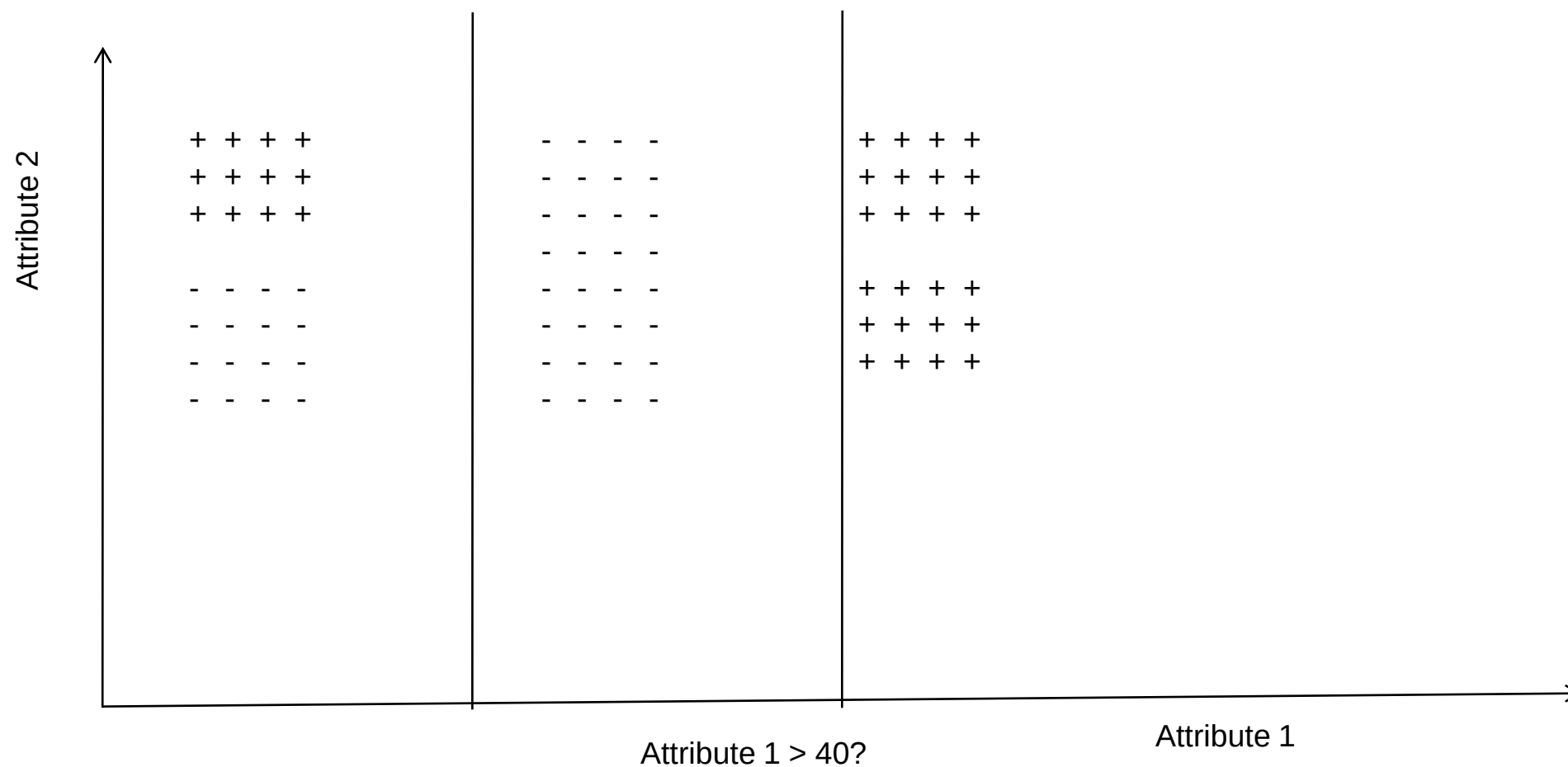
Best attribute to split?



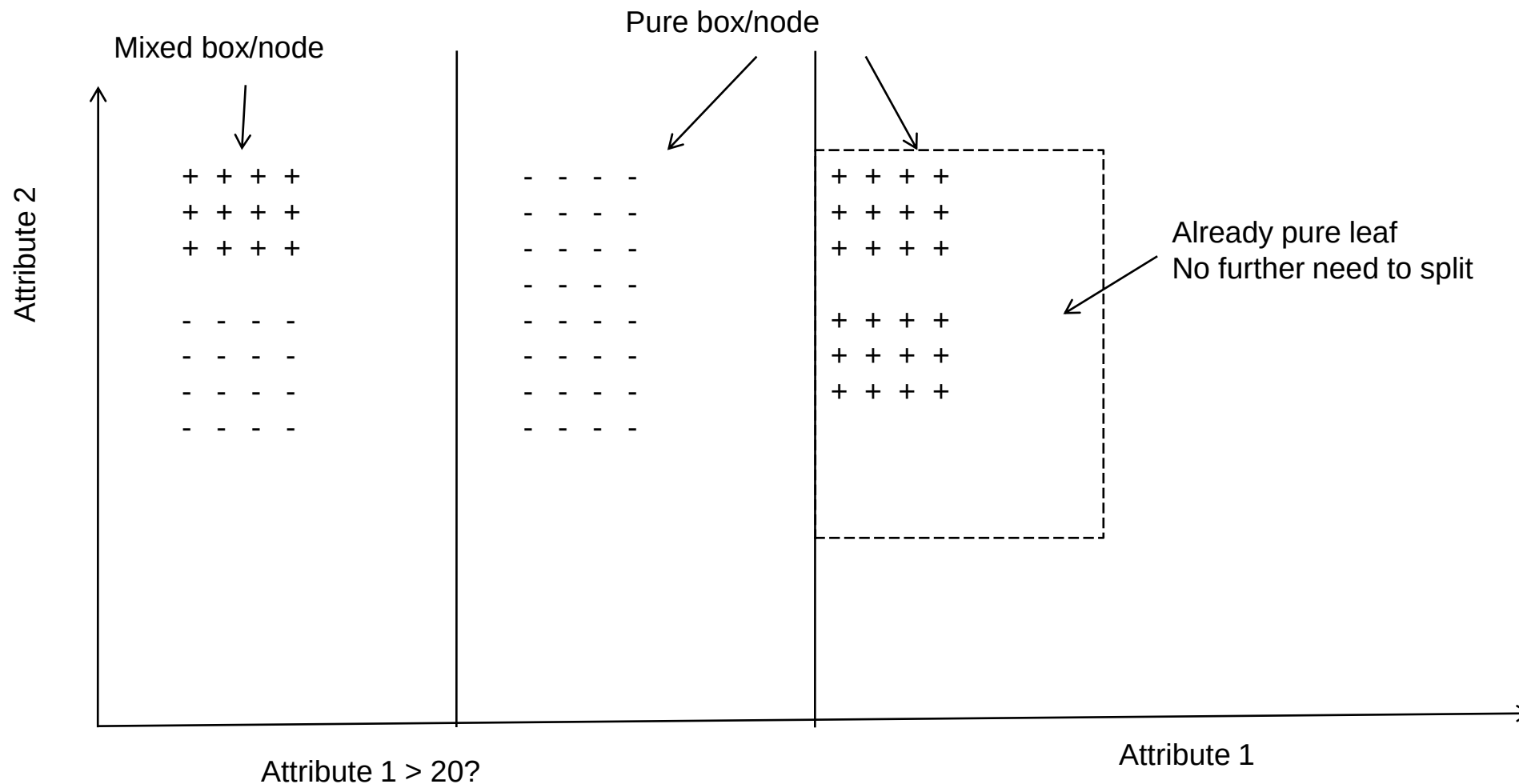
Best attribute to split?



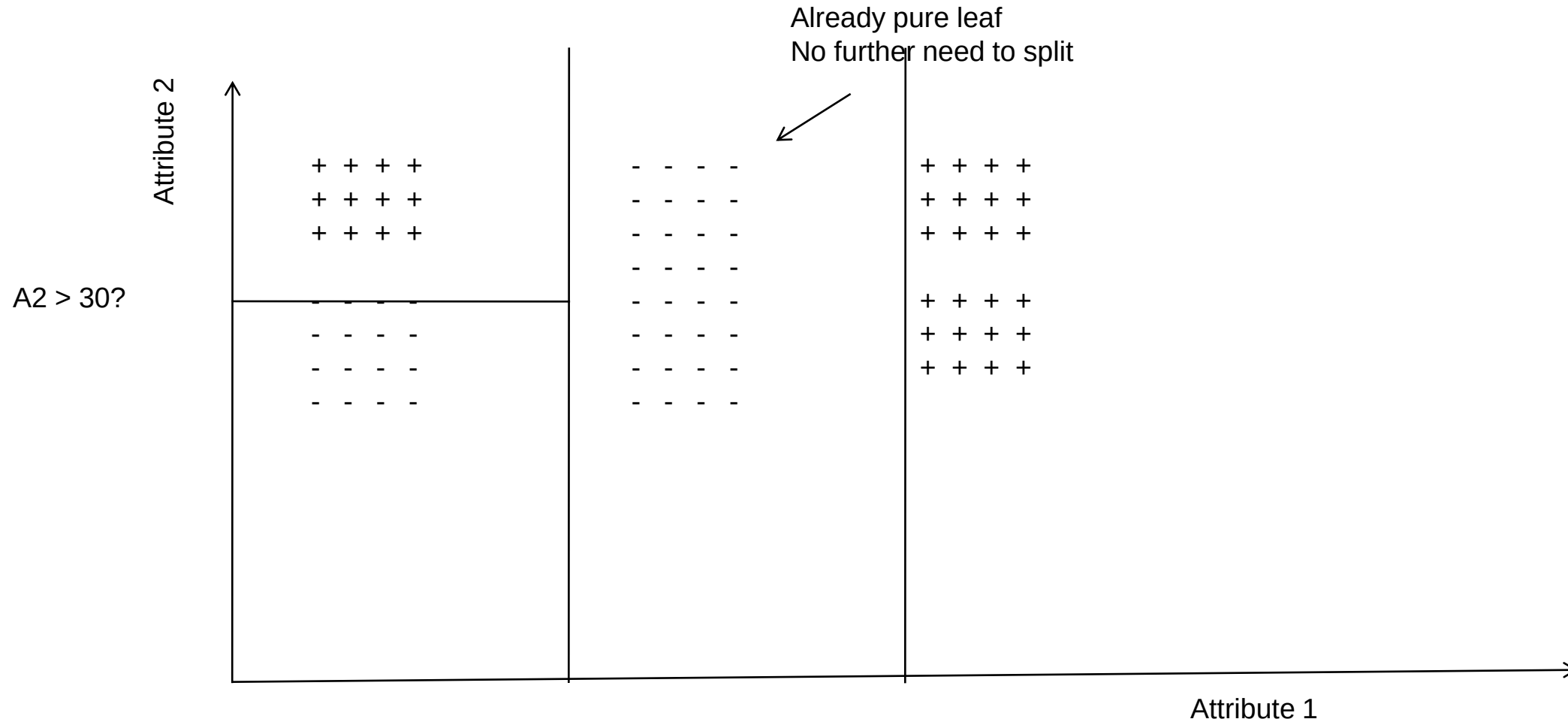
Best attribute to split?



Which split to make next?

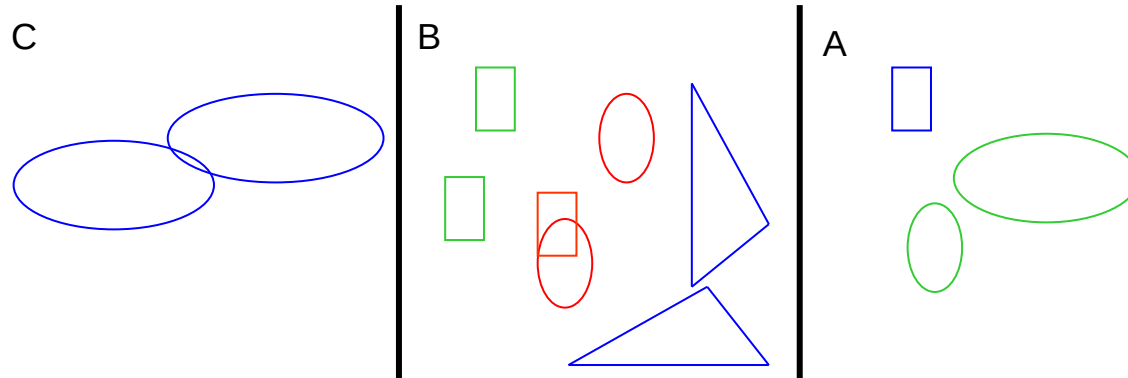


Which split to make next?



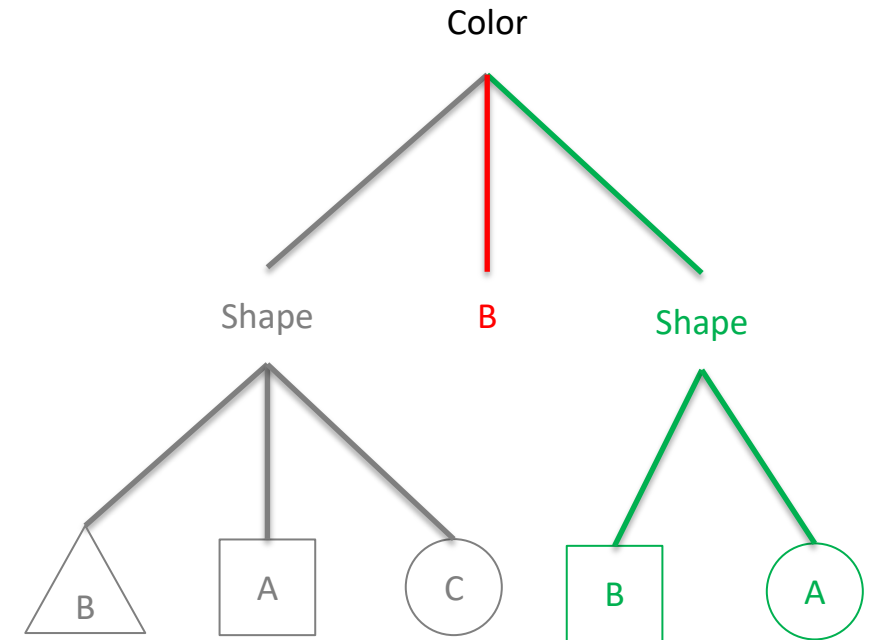
The Representation

- Decision Trees are classifiers for instances represented as feature vectors
 - color={red, blue, green} ; shape={circle, triangle, rectangle} ; label= {A, B, C}
- Nodes** are **tests** for feature values
- There is one branch for each value of the feature
- Leaves** specify the category (labels)
- Can categorize instances into multiple disjoint categories



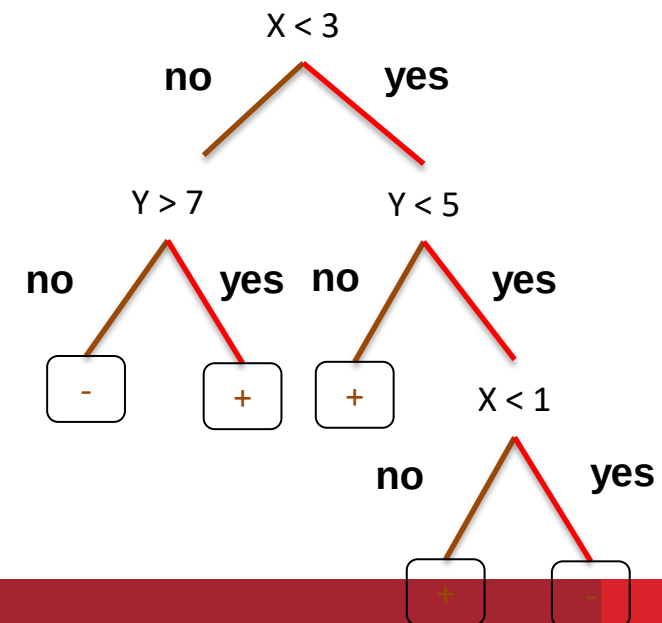
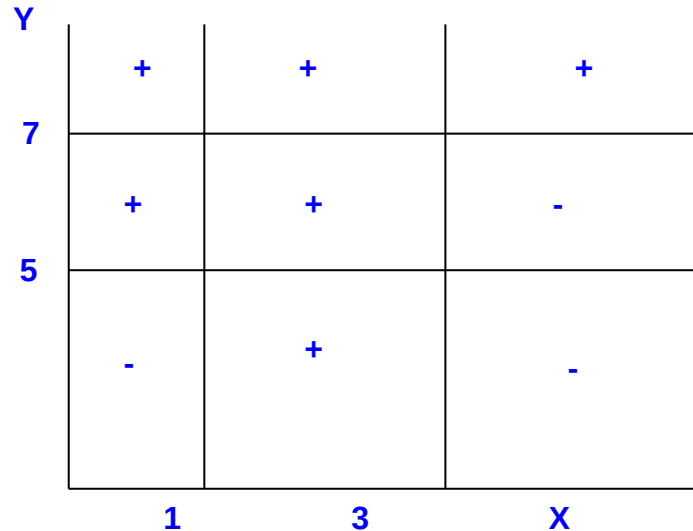
Evaluation of a
Decision Tree

Learning a
Decision Tree



Decision Boundaries

- Usually, instances are represented as attribute-value pairs (color=blue, shape = square, +)
- Numerical values can be used either by discretizing or by using thresholds for splitting nodes
- In this case, the tree divides the features space into axis-parallel rectangles, each labeled with one of the labels

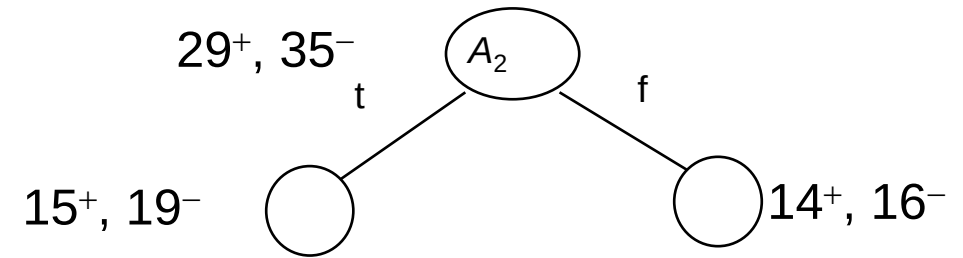
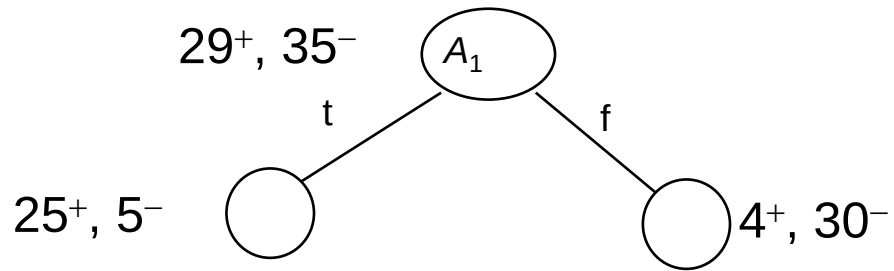


Principle of Decision Tree Construction

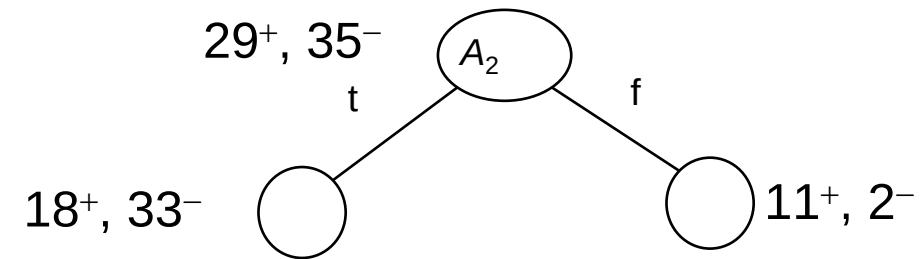
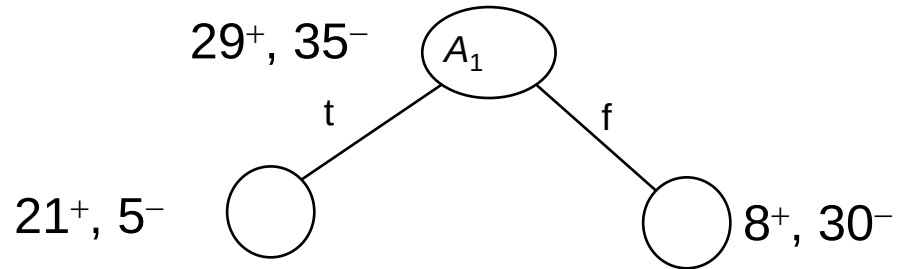
- Finally we want to form pure leaves
 - Correct classification
- Greedy approach to reach correct classification
 1. Initially treat the entire data set as a single box
 2. For each box choose the split that reduces its impurity (in terms of class labels) by the maximum amount
 3. Split the box having highest reduction in impurity
 4. Continue to Step 2
 5. Stop when all boxes are pure

Choosing Best Attribute?

- Consider 64 examples, 29^+ and 35^-
- Which one is better?



- Which is better?



Entropy

- A measure for
 - uncertainty
 - purity
 - information content
- Information theory: optimal length code assigns $(-\log_2 p)$ bits to message having probability p
- S is a sample of training examples
 - p_+ is the proportion of positive examples in S
 - p_- is the proportion of negative examples in S
- Entropy of S : average optimal number of bits to encode information about certainty/uncertainty about S
$$\text{Entropy}(S) = p_+(-\log_2 p_+) + p_-(-\log_2 p_-) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$
- Can be generalized to more than two values

Information Gain (ID3)

- The information gain of an attribute **a** is the expected reduction in entropy caused by partitioning on this attribute

$$Gain(S, a) = Entropy(S) - \sum_{v \in values(S)} \frac{|S_v|}{|S|} Entropy(S_v)$$

- Where:
 - S_v is the subset of **S** for which attribute **a** has value **v**, and
 - the entropy of partitioning the data is calculated by **weighing the entropy of each partition** by its size relative to the original set
- Partitions of low entropy (imbalanced splits) lead to high gain
- Go back to check which of the A, B splits is better



ID3

- **Iterative Dichotomiser 3 (ID3) Algorithm**
- **ID3 (Iterative Dichotomiser 3)** is a classic **decision tree algorithm** used for **classification**.
- Introduced by **Ross Quinlan** in **1986**.
- One of the **earliest and most influential** algorithms in machine learning.
- Forms the **foundation** for advanced decision tree algorithms like **C4.5** and **CART**.

ID3 (Contd.)

- The primary goal of ID3 is to:
 - Build a **decision tree** from training data.
 - **Reduce uncertainty** at each step.
 - Classify data into **distinct class labels**.
 - Create **interpretable decision rules**.
- Key Idea Behind ID3
 - Choose the **best attribute** to split the data.
 - “Best” means the attribute that:
 - **Maximizes Information Gain**
 - **Minimizes Entropy (uncertainty)**
 - Repeat this process **recursively** until classification is complete.

ID3 Algorithm

Input:

- Training dataset S
- Attribute set A
- Class label C

Output:

- A decision tree

1. If all instances in S belong to the same class → **return a leaf node**
2. If A is empty → **return a leaf with majority class**
3. Compute entropy of dataset S
4. Compute information gain for each attribute in A
5. Select attribute $A_{\{best\}}$ with **maximum information gain**
6. Create a decision node using $A_{\{best\}}$
7. Split dataset SSS based on values of $A_{\{best\}}$
8. Recursively apply ID3 to each subset

Termination Conditions (Stopping Criteria)

The ID3 algorithm stops recursion when any of the following conditions are met:

- All records in the subset belong to one class
- No attributes remain for further splitting
- Subset becomes empty
- Further splits do not reduce entropy

Example

- Given a training dataset with attributes: Age, Income, Student, Credit Rating, Target Class: Buys Computer (Yes / No)
- Goal: Use the ID3 algorithm to build a decision tree that predicts whether a customer will buy a computer.

Age	Income	Student	Credit rating	Buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

- Compute Entropy of the Entire Dataset

$$Entropy(S) = -\frac{9}{14}\log_2 \frac{9}{14} - \frac{5}{14}\log_2 \frac{5}{14} = 0.94$$

Attribute

- Age ≤ 30 (2 yes and 3 no)
- Age 31...40 (4 yes and 0 no)
- Age >40 (3 yes 2 no)

Age	Income	Student	Credit rating	Buys computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

Attribute Age

- Age ≤ 30 (2 yes and 3 no)
- Age 31...40 (4 yes and 0 no)
- Age >40 (3 yes 2 no)

Entropy(age) =

$$\begin{aligned}
 & 5/14 (-2/5 \log_2(2/5) - 3/5 \log_2(3/5)) + 4/14 (0) + \\
 & 5/14 (-3/5 \log_2(3/5) - 2/5 \log_2(2/5)) = \\
 & 5/14(0.9709) + 0 + 5/14(0.9709) = 0.6935
 \end{aligned}$$

- Gain(age) = $0.94 - 0.6935 = 0.2465$

Age	Income	Student	Credit rating	Buys computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

Consider Income Attribute

- $\text{income}_{\text{high}}$ (2 yes and 2 no), $\text{income}_{\text{medium}}$ (4 yes and 2 no), and $\text{income}_{\text{low}}$ (3 yes 1 no)
- $\text{Entropy}(\text{income}) = \frac{4}{14}(-\frac{2}{4}\log_2(\frac{2}{4}) - \frac{2}{4}\log_2(\frac{2}{4})) + \frac{6}{14}(-\frac{4}{6}\log_2(\frac{4}{6}) - \frac{2}{6}\log_2(\frac{2}{6})) + \frac{4}{14}(-\frac{3}{4}\log_2(\frac{3}{4}) - \frac{1}{4}\log_2(\frac{1}{4}))$
 $= \frac{4}{14}(1) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811)$
 $= 0.285714 + 0.393428 + 0.231714 = 0.9108$
- $\text{Gain}(\text{income}) = 0.94 - 0.9108 = 0.0292$

Age	Income	Student	Credit rating	Buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

Consider Student Attribute

- student_{yes} (6 yes and 1 no) and student_{no} (3 yes 4 no)
- Entropy(student) = $\frac{7}{14}(-\frac{6}{7}\log_2(\frac{6}{7}) - \frac{1}{7}\log_2(\frac{1}{7})) + \frac{7}{14}(-\frac{3}{7}\log_2(\frac{3}{7}) - \frac{4}{7}\log_2(\frac{4}{7}))$
 $= \frac{7}{14}(0.5916) + \frac{7}{14}(0.9852)$
 $= 0.2958 + 0.4926 = 0.7884$
- Gain (student) = $0.94 - 0.7884 = 0.1516$

Age	Income	Student	Credit rating	Buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

Consider Credit_Rating Attribute

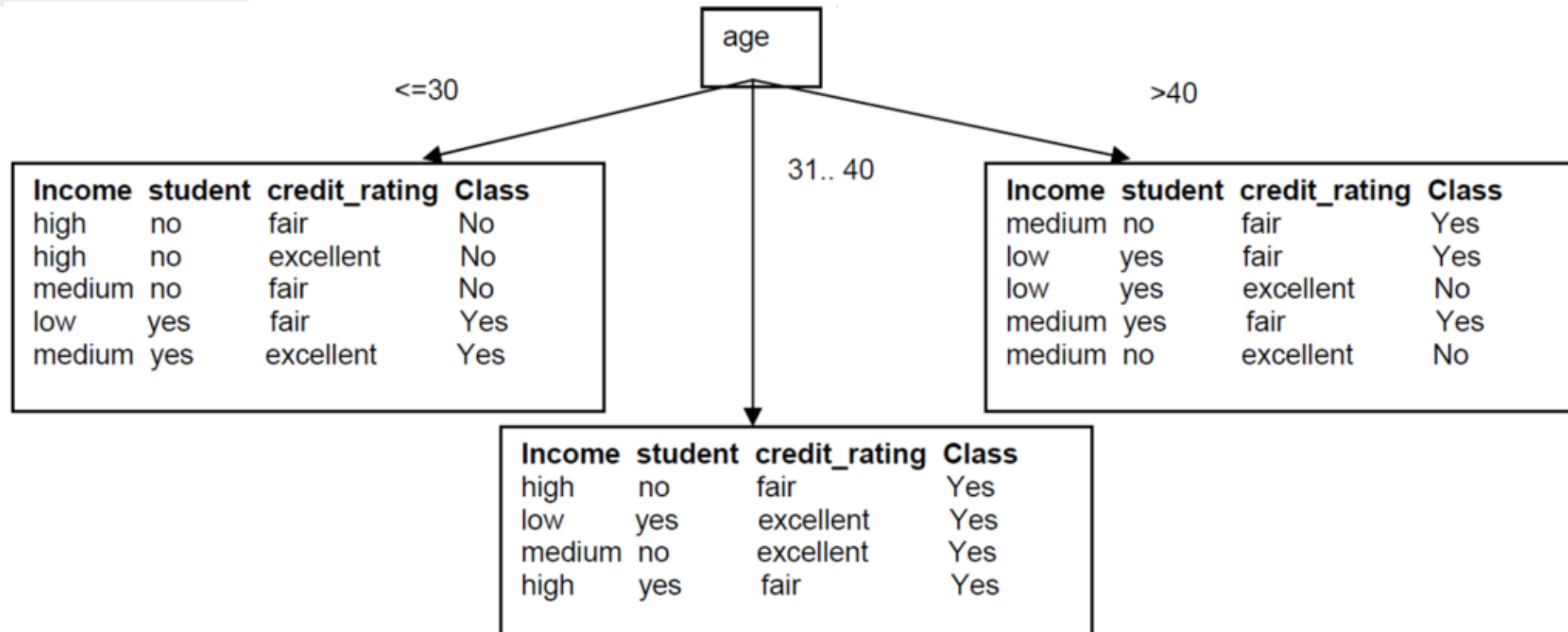
- credit_rating_fair (6 yes and 2 no) and credit_rating_excellent (3 yes 3 no)
- $\text{Entropy}(\text{credit_rating}) = \frac{8}{14}(-\frac{6}{8}\log_2(\frac{6}{8}) - \frac{2}{8}\log_2(\frac{2}{8})) + \frac{6}{14}(-\frac{3}{6}\log_2(\frac{3}{6}) - \frac{3}{6}\log_2(\frac{3}{6}))$
 $= \frac{8}{14}(0.8112) + \frac{6}{14}(1)$
 $= 0.4635 + 0.4285 = 0.8920$
- $\text{Gain}(\text{credit_rating}) = 0.94 - 0.8920 = 0.479$

Age	Income	Student	Credit rating	Buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

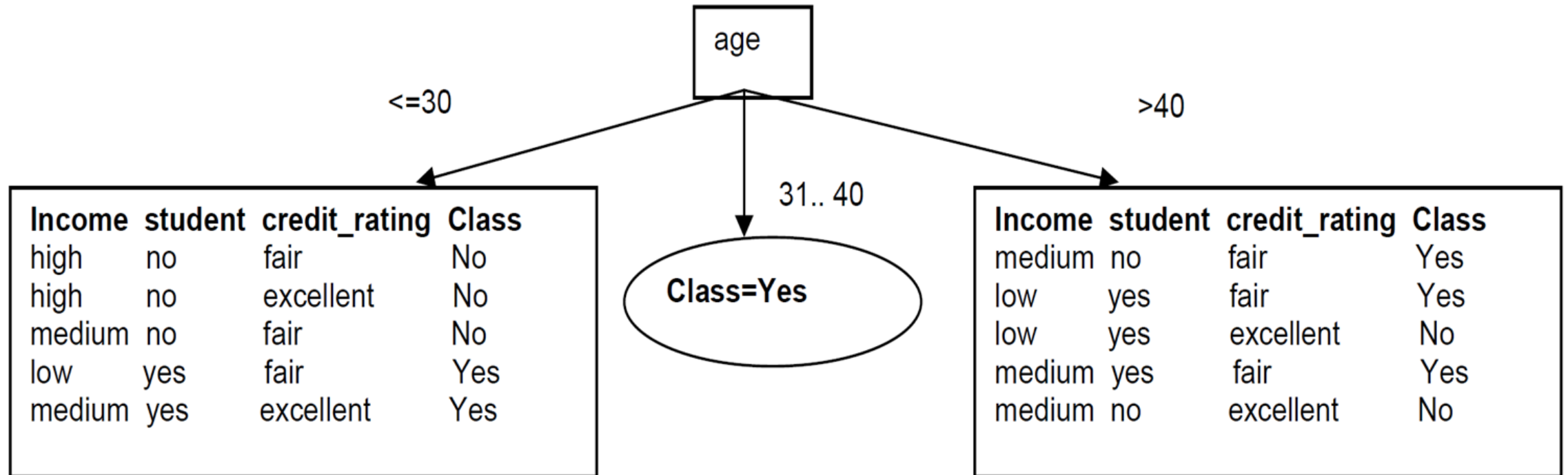
Since Age has the highest Information Gain we start splitting the dataset using the age attribute.

Attribute	Information Gain
Age	0.2465
Income	0.0292
Student	0.1516
Credit Rating	0.048



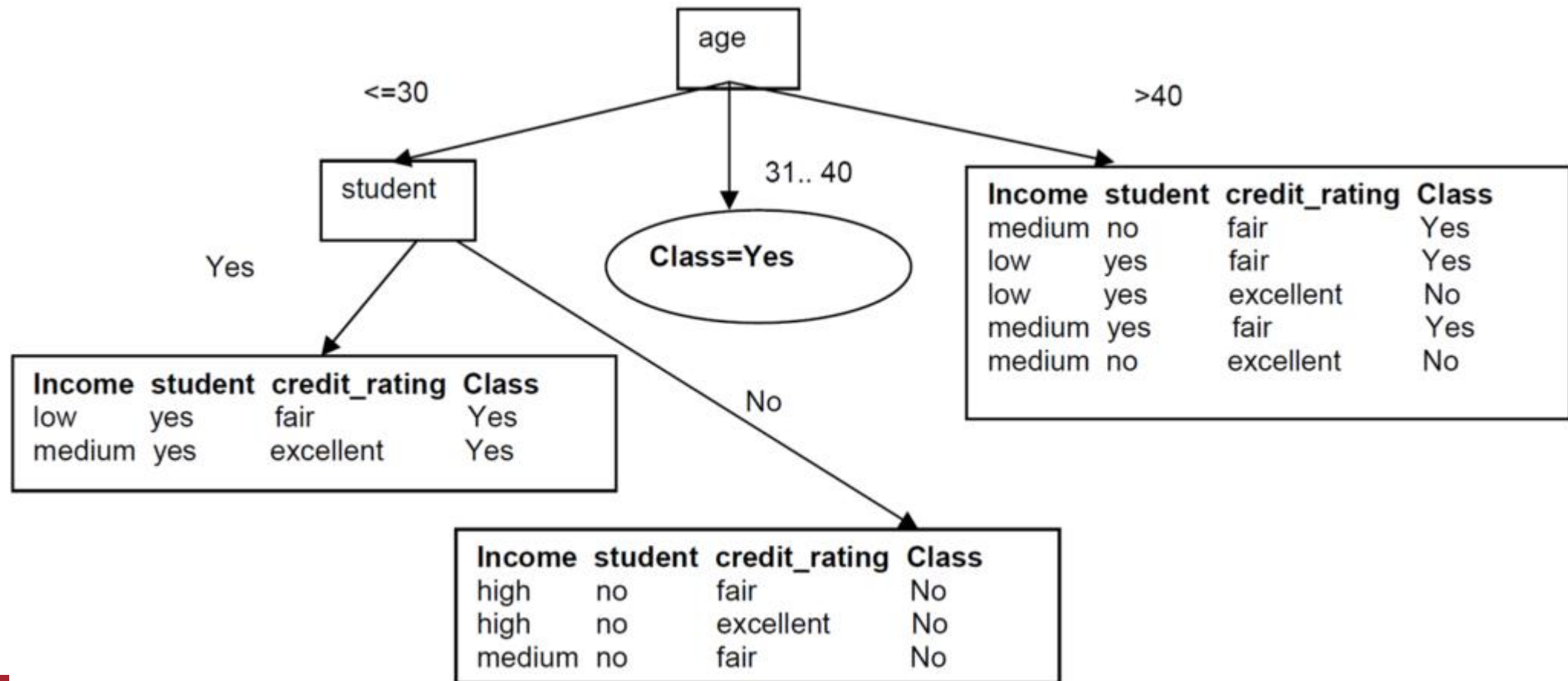
Example

- Since all records under the branch age31..40 are all of the class, Yes, we can replace the leaf with Class=Yes



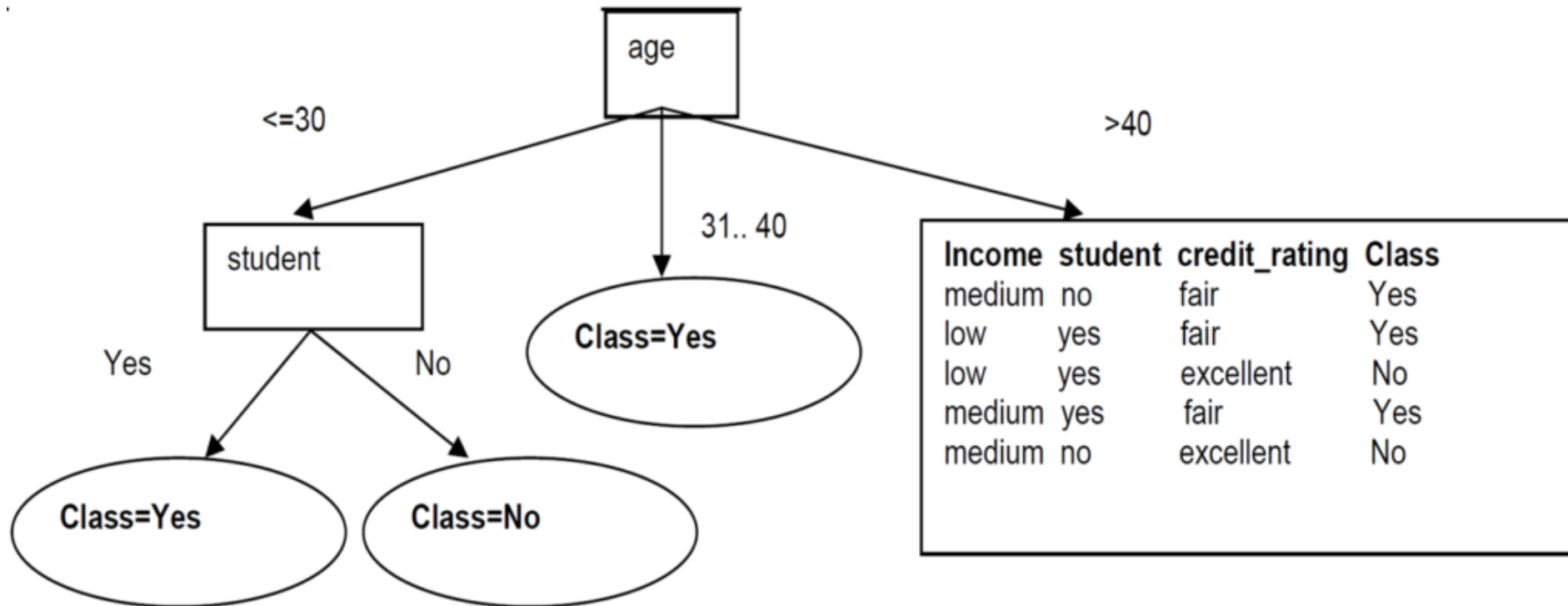
Example

- Based Income

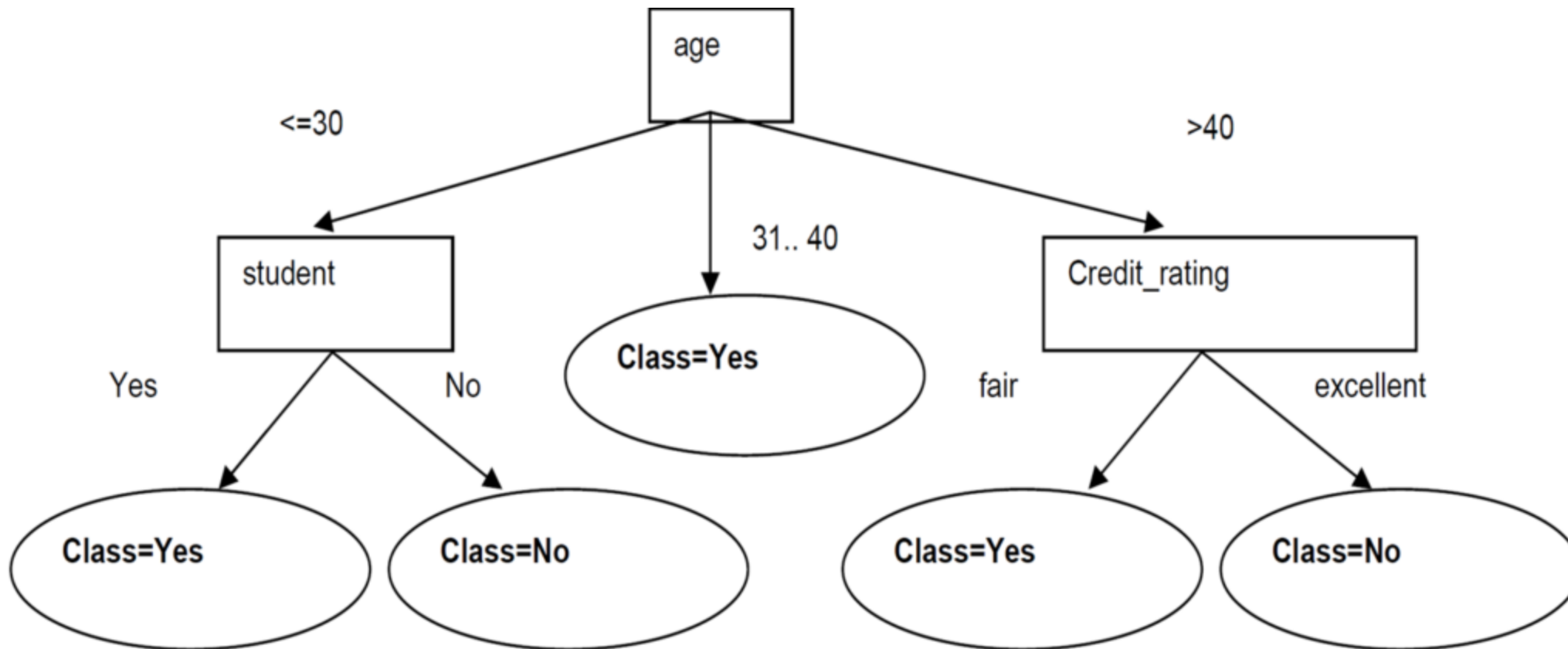


Example

- Since these two new branches are from distinct classes, we make them into leaf nodes with their respective class as label:



Final Answer



ID3 Algorithm

- **Advantages:**
 - Simple, interpretable
 - Effective for categorical data
 - Fast construction
 - Foundation for advanced algorithms
- **Limitations:**
 - Overfitting
 - Poor handling of continuous data
 - Bias toward multi-valued attributes
 - No pruning
 - Scalability issues

Gain Ratio for Attribute Selection (C4.5)

- Information gain measure is biased towards attributes with a large number of values
- C4.5 (a successor of ID3) uses gain ratio to overcome the problem (normalization to information gain)

$$SplitInfo_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

$$- \text{GainRatio}(A) = \text{Gain}(A) / \text{SplitInfo}(A)$$

Ex. $SplitInfo_{income}(D) = -\frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) - \frac{6}{14} \times \log_2 \left(\frac{6}{14} \right) - \frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) = 1.557$

$$- \text{gain_ratio}(\text{income}) = 0.029 / 1.557 = 0.019$$

- The attribute with the maximum gain ratio is selected as the splitting attribute

Reference

- <https://www.mygreatlearning.com/blog/decision-tree-algorithm/>
- <https://www.upgrad.com/blog/decision-tree-in-ai/>